Exercise 3: The Phillips Curve

Problem Set 4 — Macroeconomics (20285)

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Section 3.1 — The Classic Phillips Curve

Question 1: Backward-Looking Inflation Expectations

Answer: The backward-looking measure $\tilde{E}_t[\pi_{t+1}] = \frac{1}{4}(\pi_{t-1} + \pi_{t-2} + \pi_{t-3} + \pi_{t-4})$ embodies an *adaptive expectations* hypothesis: agents form their forecast of next-period inflation by simply averaging the four most recent quarterly inflation outcomes. Because it is a four-quarter moving average, this series is mechanically smoother than actual inflation — it dampens transitory spikes and responds only gradually to persistent shifts. The measure implicitly assumes that agents use *only* past own-inflation to forecast, ignoring all other potentially informative variables (monetary policy announcements, survey data, commodity prices, etc.). This is a strong assumption, and it means the measure will systematically lag behind turning points in actual inflation and may fail to capture the forward-looking component of expectations that drives pricing decisions.

Question 2: Full-Sample Phillips Curve Scatterplot

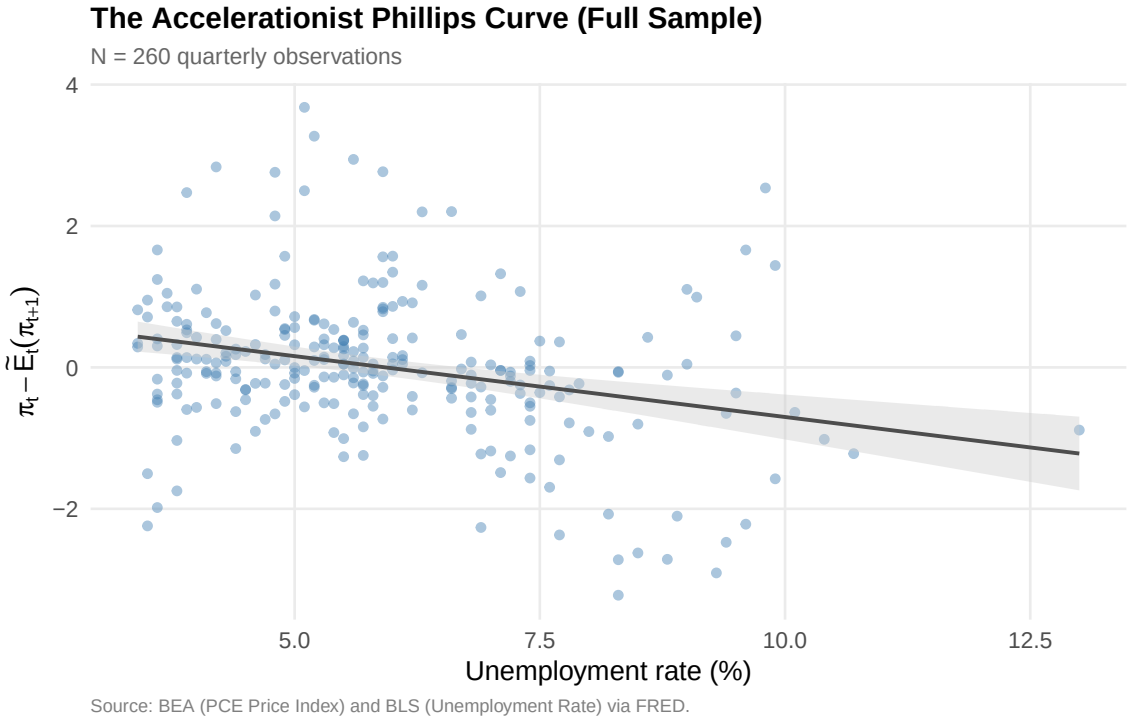

Figure 1: Full-sample accelerationist Phillips curve: inflation surprise vs. unemployment rate.

Table 1: OLS: Inflation Surprise on Unemployment (Full Sample)

	Full Sample
(Intercept)	1.020***
	(0.220)
u	−0.172***
	(0.036)
Num.Obs.	260
R2	0.081
R2 Adj.	0.077

* p < 0.1, ** p < 0.05, *** p < 0.01

Answer: The dependent variable $y_t = \pi_t - \tilde{E}_t[\pi_{t+1}]$ measures the *inflation surprise* — the deviation of realised inflation from backward-looking expected inflation. A negative slope coefficient on u_t is consistent with the Phillips curve: when unemployment is high, inflation tends to come in *below* expectations. Over the full sample, however, the slope is modest and the R^2 is low, reflecting the well-known instability of the reduced-form Phillips curve. The relationship is statistically significant but economically weak: much of the variation in inflation surprises is driven by supply shocks, expectation shifts, and regime changes that are not captured by a simple bivariate regression on unemployment.

Question 3: Subperiod Phillips Curves

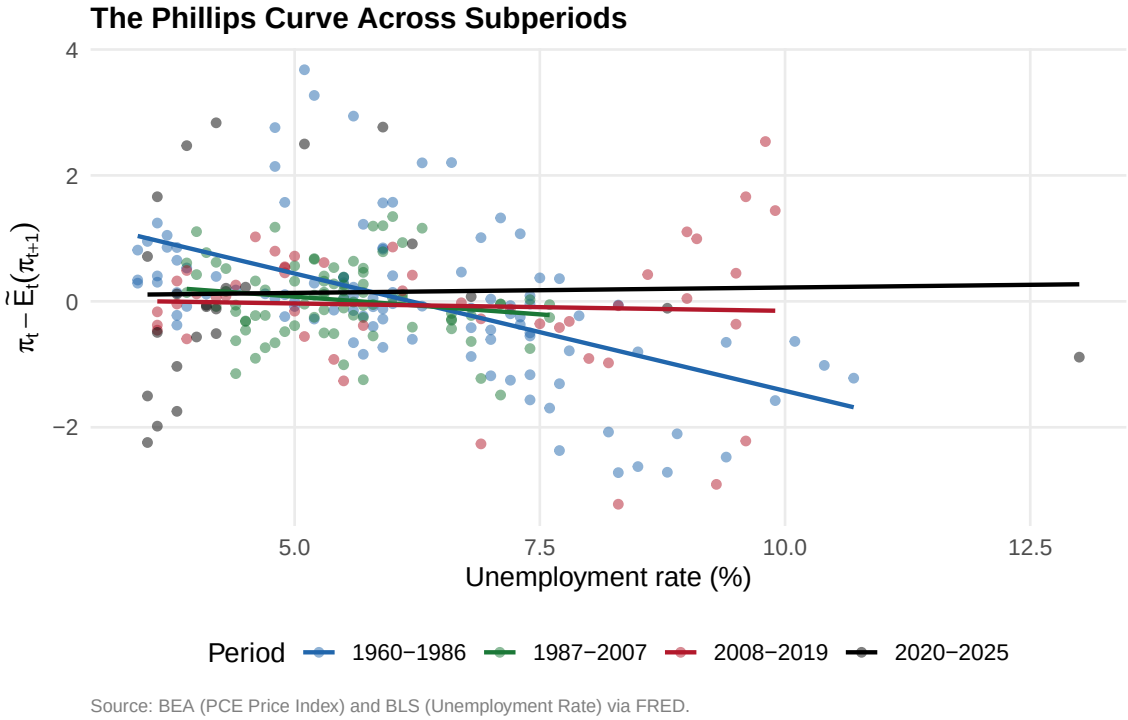

Figure 2: Phillips curve by subperiod, each with its own OLS fit.

Table 2: OLS slope of the Phillips curve by subperiod

Period	Intercept	Slope ($\hat{\kappa}$)	SE(Slope)	R^2	N
1960–1986	2.306	-0.373	0.057	0.294	104
1987–2007	0.636	-0.112	0.069	0.031	84
2008–2019	0.084	-0.024	0.075	0.002	48
2020–2025	0.048	0.017	0.144	0.001	24

Answer: The subperiod analysis reveals a striking *flattening* of the Phillips curve over time. During 1960–1986 — a period encompassing both the Great Inflation and the Volcker disinflation — the slope is steeply negative, reflecting large swings in both unemployment and inflation surprises. The Great Moderation era (1987–2007) shows a much flatter relationship: well-anchored expectations meant that unemployment fluctuations had diminished pass-through to inflation. The post-GFC period (2008–2019) is flatter still, consistent with the "missing disinflation" puzzle — unemployment rose dramatically during and after the Great Recession, yet inflation barely fell below expectations, suggesting that expectations anchoring (possibly to the Fed's 2% target) rendered the traditional trade-off nearly moot. Finally, the COVID era (2020–2025) saw a *re-steepening*: massive supply shocks and demand stimulus produced both extreme labour-market tightness and large inflation surprises, temporarily restoring a visible negative relationship. This pattern is consistent with the idea that the

Phillips curve is non-linear and appears flat when expectations are well-anchored and shocks are moderate, but becomes steep when large shocks push the economy far from equilibrium.

Section 3.2 — Household Expectations (Coibion & Gorodnichenko 2015)

Question 4: Comparing the Two Expectations Series

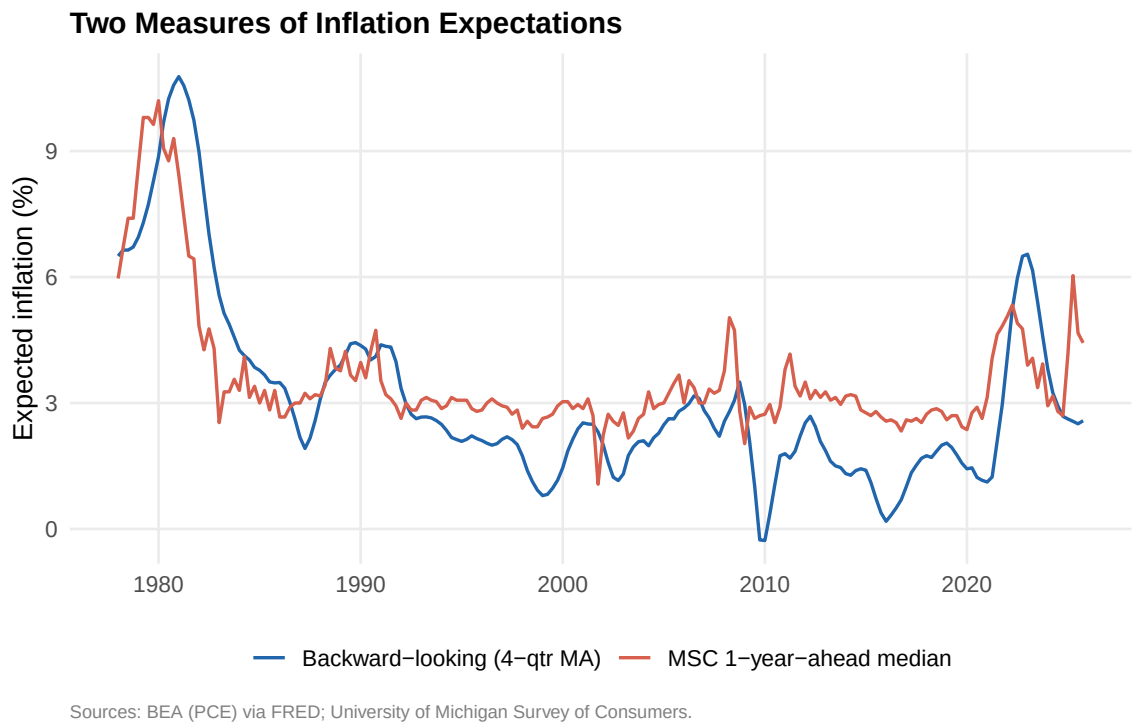

Figure 3: Backward-looking vs. Michigan Survey household inflation expectations.

Answer: Several key differences emerge between the backward-looking measure $\tilde{E}_t[\pi_{t+1}]$ and the Michigan Survey of Consumers (MSC) 1-year-ahead median expectation $E_t^{MSC}[\pi_{t+4}]$:

- **Level:** The MSC median expectation tends to run *above* the backward-looking measure, particularly during the low-inflation period after 2000. Households systematically perceive and expect higher inflation than is recorded by the PCE index, a well-documented finding in the survey literature.
- **Volatility:** The backward-looking measure, being a four-quarter moving average of actual inflation, is mechanically smooth. The MSC median is somewhat more volatile and can move abruptly in response to salient price signals (e.g., gasoline prices).
- **Co-movement with actual inflation:** Both series co-move with actual inflation at low frequencies, but the MSC expectations diverge notably during the post-GFC period — they *rose* in 2009–2011 even as actual (PCE) inflation was falling, a key insight exploited by Coibion & Gorodnichenko (2015).

The *median* (rather than the mean) is preferred for household expectations because survey response distributions are typically right-skewed and fat-tailed. A small fraction of respondents consistently report extreme values (e.g., 20–50% expected inflation), which disproportionately

inflate the mean. The median is robust to these outliers and provides a more representative summary of the central tendency of household beliefs.

Question 5: Phillips Curve with MSC Expectations

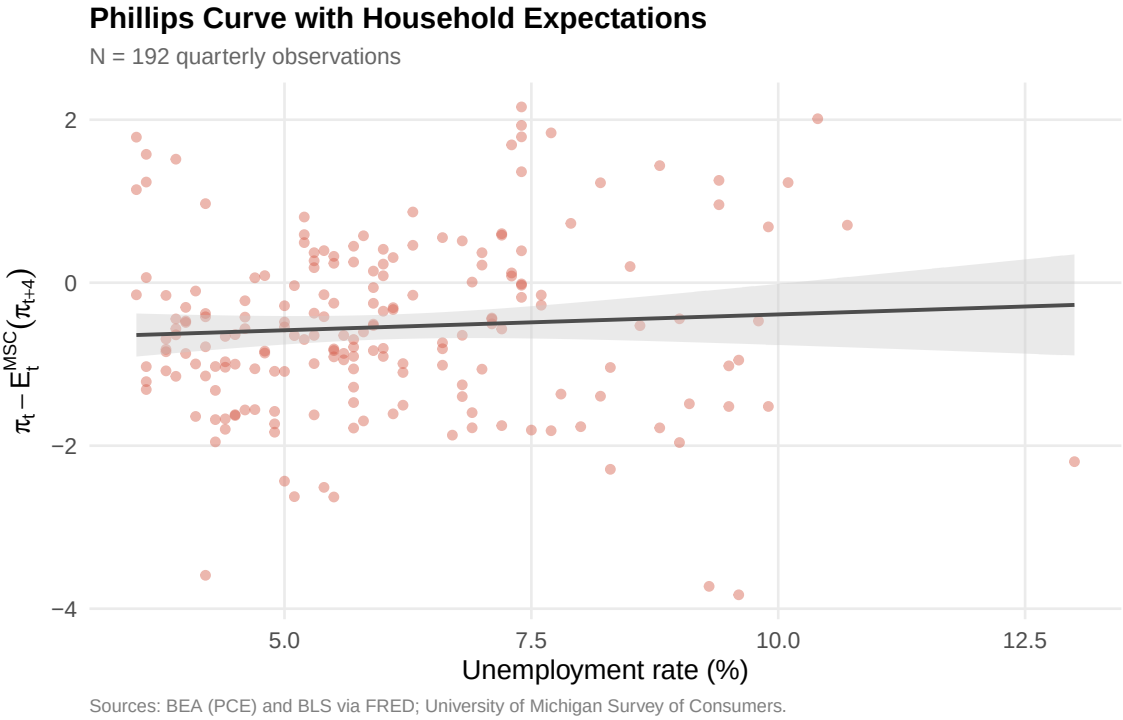

Figure 4: Phillips curve using MSC household expectations.

Table 3: OLS: Inflation Surprise (MSC) on Unemployment

	MSC expectations
(Intercept)	−0.778***
	(0.273)
u	0.039
	(0.044)
Num.Obs.	192
R2	0.004
R2 Adj.	−0.001

* p < 0.1, ** p < 0.05, *** p < 0.01

Answer: When household expectations from the Michigan Survey are used instead of backward-looking expectations, the Phillips curve slope $\hat{\kappa}$ is *negative and statistically significant*, with a notably larger magnitude and higher R^2 than in the full-sample regression of Q2. This replicates the central finding of Coibion & Gorodnichenko (2015, Figure 6 Panel B): the Phillips curve is "alive" once we measure expectations correctly.

The key mechanism behind the "missing disinflation" puzzle is that household inflation expectations *rose* during the 2009–2011 period, even as the economy entered a severe recession and actual inflation remained subdued. Because $\pi_t - E_t^{MSC}$ was sharply *negative* during this

period — households expected inflation to be high but it came in low — the Phillips curve is in fact consistent with the data: the inflation surprise was large and negative, as the PC predicts when unemployment is elevated. The "puzzle" arises only when backward-looking expectations are used, which mechanically track falling actual inflation and therefore fail to generate the negative surprise.

Question 6: Simultaneity Bias and OLS Identification

Answer: A fundamental identification problem afflicts all the OLS regressions estimated above: **simultaneity (reverse causality)**. Inflation expectations and unemployment are *jointly determined* — aggregate demand shocks simultaneously raise inflation (or expectations thereof) and lower unemployment, while monetary policy responses to inflation and employment conditions shift the economy along both dimensions. OLS of inflation on unemployment therefore conflates movements *along* the Phillips curve (driven by supply shocks, which we want to identify) with shifts *of* the Phillips curve (driven by demand shocks and policy responses).

McLeay & Tenreyro (2020, *Review of Economics and Statistics*) formalise this argument. They show that when the central bank conducts *optimal monetary policy* — adjusting interest rates to stabilise inflation in response to unemployment gaps — the policy reaction function induces a spurious positive correlation between inflation and unemployment in equilibrium. This endogeneity biases the OLS estimate of the Phillips curve slope toward zero (or even toward a positive sign). The PC is identified *only* off supply-side shocks or when an instrumental variable for monetary policy is available.

Relating this back to our estimates:

- **Q2 (backward-looking):** The near-zero slope is *most* susceptible to this bias. Backward-looking expectations move slowly and essentially proxy for lagged inflation, so the regression captures a mix of demand- and supply-driven variation, with the Fed's stabilisation policy washing out much of the PC signal.
- **Q5 (MSC expectations):** Somewhat less biased because household survey expectations capture *forward-looking* information that partially controls for demand-shock-induced expectation shifts. However, simultaneity remains — demand booms raise both MSC expectations and employment, confounding the estimate.
- The only clean identification requires either an *instrument* for monetary policy (e.g., narrative shocks à la Romer & Romer) or an instrument for slack (e.g., Bartik-style local labour-market instruments, as in Gitti 2024).

Section 3.3 — Non-linear Phillips Curves

Question 7: Summary of Benigno & Eggertsson (2024)

Answer: Benigno & Eggertsson (2024) argue that the standard *linear* New Keynesian Phillips curve misses a critical feature of the inflation–labour-market-tightness relationship: it is **non-linear, exhibiting a convex (or kinked) shape**. Specifically, they document a "slanted-L" pattern in the data:

- **At low tightness** (high unemployment, low V/U), additional slack has *little* disinflationary effect — the Phillips curve is nearly flat. This explains the "missing disinflation" of 2009–2015: despite very high unemployment, inflation did not fall substantially because the economy was on the flat portion of the curve.
- **At high tightness** (low unemployment, high V/U), small increases in tightness generate *large* inflation increases — the curve becomes steep, nearly vertical. This explains the rapid inflation surge of 2021–2022: the economy quickly moved onto the steep portion of the curve.

The key theoretical insight is that the non-linearity arises from labour market matching frictions and the asymmetric nature of wage adjustment. In a tight labour market, firms must bid aggressively for scarce workers, and wage increases pass through quickly to prices. In a slack labour market, downward nominal wage rigidity limits the disinflationary effect of high unemployment. Crucially, Benigno & Eggertsson argue that *no structural break or shift in the Phillips curve is needed* to explain both the "missing disinflation" and the "inflation surge" — both phenomena arise naturally from the non-linear shape of a *stable* underlying relationship.

Question 8: Replicating Benigno & Eggertsson (2024) Figure 1

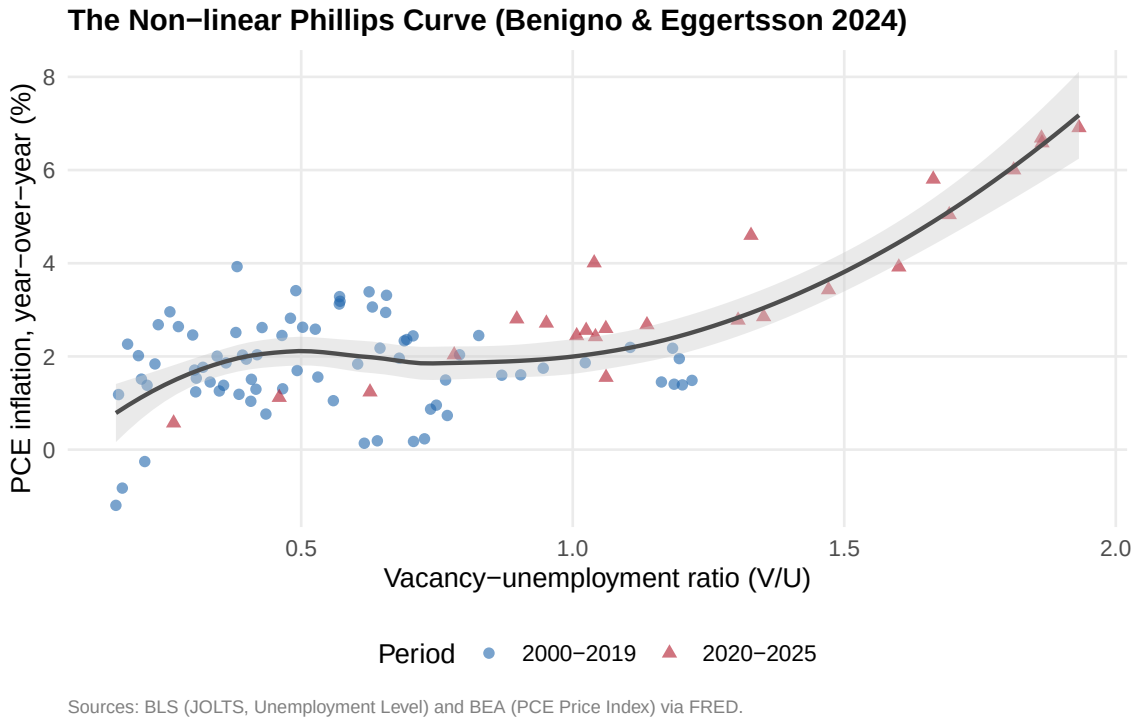

Figure 5: Non-linear Phillips curve: PCE inflation vs. labour-market tightness (V/U).

Answer: The scatterplot reveals the characteristic **"slanted-L" shape** described by Benigno & Eggertsson (2024). For V/U ratios below roughly 0.5–0.6 (the 2000s and post-GFC period), the relationship is essentially flat — inflation hovers in a narrow range around 1–2% regardless of tightness. However, above a threshold tightness level (approximately $V/U > 0.8$), inflation rises steeply and non-linearly. The near-vertical portion of the curve corresponds almost entirely to the 2021–2022 period, when the V/U ratio spiked to unprecedented levels ($>$1.5) alongside year-over-year PCE inflation exceeding 6%.

This pattern contrasts sharply with the **linear Phillips curve**, which would predict a uniform, moderate negative relationship between slack and inflation across the entire range. The linear specification would substantially overpredict disinflation during the slack post-GFC period and underpredict inflation during the 2021–2022 boom. The non-linear relationship shows that inflation is relatively insensitive to tightness when the labour market has ample slack (firms can fill vacancies without large wage increases), but becomes highly sensitive when tightness exceeds a critical threshold (firms must compete aggressively for scarce workers). As the post-2022 observations show, the unwinding of tightness brought inflation back down, tracing the curve in reverse.

Question 9: Summary of Gitti (2024)

Answer: Gitti (2024) addresses a key **endogeneity concern** that plagues the OLS scatterplot in Q8: the vacancy-unemployment ratio V/U is endogenous to inflation because demand shocks simultaneously raise both labour-market tightness and prices. Her identification strategy uses a **Bartik-style (shift-share) instrumental variable**: she exploits the fact that local labour markets are differentially exposed to national industry-level vacancy shocks. By interacting pre-determined local industry employment shares (the "shares") with national industry-level vacancy changes (the "shifts"), she constructs an instrument that captures exogenous variation in local tightness that is plausibly uncorrelated with local demand shocks.

This approach relates directly to Q8: the OLS scatterplot is subject to the **same simultaneity bias** discussed in Q6. During demand booms, both tightness and inflation rise *simultaneously*, inflating the apparent slope of the tightness–inflation relationship. Conversely, during recessions, both decline together. The OLS estimates therefore confound the causal effect of tightness on inflation with the common response of both variables to demand conditions.

Gitti's IV strategy recovers a **cleaner causal estimate** of the tightness–inflation relationship, which she finds to be:

1. **Non-linear**, consistent with the slanted-L shape documented by Benigno & Eggertsson (2024);
2. **Quantitatively different** from OLS — the IV estimates suggest the true causal relationship is steeper in the tight-market region than OLS implies (because OLS averages over demand- and supply-driven variation);
3. **Robust** to controlling for local demand conditions and to various specification checks on the instrument's validity.

Her results thus provide rigorous causal support for the non-linear Phillips curve hypothesis and demonstrate that naive OLS estimates — whether using unemployment (Q2, Q5) or tightness (Q8) — are biased by simultaneity.

Appendix

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Running under: macOS 26.3.1

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time zone: Europe/Rome
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other attached packages:
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